

The Hive: Deep Technical Execution Report

Mechanics, Artifacts, and Detection of Emulated ATT&CK Techniques

This technical report breaks down exactly how The Hive mathematically models and technically executes Advanced Persistent Threat (APT) behavior. It details the precise commands run by the Operator Fleet, the forensic artifacts left on the target system, and the telemetry analyzed by the Purple Team Agent.

The following breakdown is based on the 12 specific techniques extracted during the APT29 (Cozy Bear) emulation profile.

I. Phase 1: Reconnaissance (Network & Host Discovery)

1. T1082 - System Information Discovery

Adversary Goal

Determine OS version, hardware configurations, and patch levels to tailor local privilege escalation exploits.

- **Hive Execution Mechanism:** The Recon Agent executes native Windows API calls and WMI queries, specifically running: systeminfo, wmic os get /format:list, and Get-ComputerInfo.
- **Forensic Footprint:** Spawning of wmic.exe and systeminfo.exe from anomalous parent processes (e.g., powershell.exe rather than explorer.exe).
- **Purple Team Detection Logic:** Generates a Sigma rule looking for rapid, successive execution of built-in discovery binaries (Event ID 4688 Process Creation) within a narrow 5-second timeframe.

2. T1018 - Remote System Discovery

Adversary Goal

Enumerate active hosts on the subnet to identify targets for lateral movement (e.g., Domain Controllers, File Servers).

- **Hive Execution Mechanism:** Uses nltest /domain_trusts, net view /domain, and active ICMP/ARP sweeps via PowerShell Test-NetConnection.
- **Forensic Footprint:** High volume of ARP requests from a single endpoint; execution of nltest.exe querying Domain Trusts.
- **Purple Team Detection Logic:** Sigma rule targeting nltest.exe command-line arguments specifically querying trust relationships, which is highly unusual for standard user behavior.

3. T1016 - System Network Configuration Discovery

Adversary Goal

Identify network topology, routing tables, and active network connections.

- **Hive Execution Mechanism:** The agent runs ipconfig /all, arp -a, and route print.
- **Forensic Footprint:** Execution of these utilities in quick succession.
- **Purple Team Detection Logic:** D3FEND mapping to Network Traffic Analysis; looking for arp.exe execution combined with outbound network scanning behavior.

II. Phase 2: Initial Access & Execution

4. T1566.001 - Spearphishing Attachment

Adversary Goal

Trick a user into opening a malicious document, establishing the initial foothold.

- **Hive Execution Mechanism:** The Access Agent drops an inert, macro-enabled Word document (.docm) or an LNK shortcut disguised as a PDF onto the file system, then simulates a user click via ShellExecute.
- **Forensic Footprint:** WINWORD.EXE spawning child processes like cmd.exe or powershell.exe.
- **Purple Team Detection Logic:** Hard Sigma rule targeting Microsoft Office applications acting as parent processes to command-line interpreters (Suspicious Process Lineage).

5. T1059.001 - Command and Scripting Interpreter: PowerShell

Adversary Goal

Execute malicious payloads natively in memory without touching the disk (Fileless execution).

- **Hive Execution Mechanism:** Executes a Base64 encoded payload: powershell.exe -NoProfile -ExecutionPolicy Bypass -WindowStyle Hidden -EncodedCommand <BASE64_STRING>
- **Forensic Footprint:** PowerShell Script Block Logging (Event ID 4104) capturing the decoded script; powershell.exe running with hidden window arguments.
- **Purple Team Detection Logic:** YARA rule targeting the Base64 decoding stubs; Sigma rule flagging ExecutionPolicy Bypass and -EncodedCommand flags.

6. T1053.005 - Scheduled Task

Adversary Goal

Execute code at a specific time or under a specific system context (SYSTEM privileges).

- **Hive Execution Mechanism:** The agent registers a task via the schtasks.exe binary: schtasks /create /tn "WindowsUpdateCore" /tr "C:\Temp\payload.exe" /sc onstart /ru SYSTEM
- **Forensic Footprint:** Windows Security Event ID 4698 (A scheduled task was created).
- **Purple Team Detection Logic:** Rule alerting on any Scheduled Task creation where the binary path points to C:\Users* or C:\Temp* instead of C:\Windows\System32.

7. T1027 - Obfuscated Files or Information

Adversary Goal

Evade signature-based antivirus detection by hiding the true nature of the payload.

- **Hive Execution Mechanism:** The payload is XOR-encrypted or packed using a custom stub before being written to disk or memory.
- **Forensic Footprint:** High entropy files written to disk; AMSI (Antimalware Scan Interface) bypass attempts in memory.
- **Purple Team Detection Logic:** YARA rule analyzing file entropy; if entropy exceeds 7.2 (highly packed/encrypted), it flags the file for dynamic analysis.

8. T1070.004 - File Deletion

Adversary Goal

Remove forensic evidence (droppers, configuration files) to hinder incident response.

- **Hive Execution Mechanism:** Uses Remove-Item -Force or native SDelete (Sysinternals) to wipe staging directories.
- **Forensic Footprint:** Sysmon Event ID 11 (File Create) followed rapidly by Sysmon Event ID 23 (File Delete) or Event ID 26 (File Delete logged).
- **Purple Team Detection Logic:** Alerts on the execution of sdelete.exe or rapid creation-and-deletion cycles of executable files in temporary directories.

III. Phase 3: Persistence

9. T1547.001 - Registry Run Keys / Startup Folder

Adversary Goal

Ensure the malware automatically executes every time the machine reboots or the user logs in.

- **Hive Execution Mechanism:** The Persistence Agent executes: reg add "HKCU\Software\Microsoft\Windows\CurrentVersion\Run" /v "OneDriveSync" /t REG_SZ /d "C:\Users\Public\payload.exe" /f
- **Forensic Footprint:** Sysmon Event ID 13 (Registry Event - Value Set) modifying the Run or RunOnce keys.
- **Purple Team Detection Logic:** High-confidence Sigma rule alerting on any modification to HKCU\Software\Microsoft\Windows\CurrentVersion\Run by non-system processes like powershell.exe or cmd.exe.

IV. Phase 4: Lateral Movement

10. T1550.002 - Pass the Hash

Adversary Goal

Authenticate to remote servers without knowing the plaintext password, using a stolen NTLM hash.

- **Hive Execution Mechanism:** The Movement Agent simulates Mimikatz-style PtH by injecting an NTLM hash into the LSASS process and requesting a Kerberos TGT or NTLM authentication to \\DC01\C\$.
- **Forensic Footprint:** Windows Security Event ID 4624 (Logon) with Logon Type 9 (NewCredentials) matching the stolen account, followed by SMB connections.
- **Purple Team Detection Logic:** Detects Event ID 4624 (Logon Type 9) originating from abnormal processes; monitors lsass.exe memory access (Sysmon Event ID 10) by suspicious binaries.

11. T1021.002 - SMB/Windows Admin Shares

Adversary Goal

Move binary payloads to remote servers and execute them via native administrative protocols.

- **Hive Execution Mechanism:** Copies a payload over port 445: copy payload.exe \\FS01\C\$\Windows\Temp\ and executes it remotely via WMI or PsExec.
- **Forensic Footprint:** Sysmon Event ID 3 (Network Connection) on port 445; Event ID 5140 (A network share object was accessed) specifically targeting ADMIN\$ or C\$.
- **Purple Team Detection Logic:** Alerts on workstation-to-workstation SMB traffic over port 445 (standard environments should only see workstation-to-server traffic).

V. Phase 5: Exfiltration

12. T1048.002 - Exfiltration Over Asymmetric Encrypted Non-C2 Protocol

Adversary Goal

Steal sensitive data while blending in with normal corporate web traffic to bypass DPI (Deep Packet Inspection).

- **Hive Execution Mechanism:** The Exfil Agent collects files into a password-protected .zip archive, then pushes the data out via HTTPS (Port 443) to a cloud storage provider (e.g., AWS S3, Mega, or a custom VPS).
- **Forensic Footprint:** Large, sustained outbound network connections over port 443 from non-browser processes (e.g., curl.exe or powershell.exe).
- **Purple Team Detection Logic:** Sigma rule looking for byte transfers exceeding 50MB in a single HTTPS session originating from anomalous processes. D3FEND mapping points to implementing strict outbound proxy filtering.